[B1] – correct shape
[B1] – label 8.8 Mev and 56 at the peak of graph

**(c) (i)** [B1] – all 3 positions marked in correct sequence, relative spacing is correct

**(ii)** Binding energy per nucleon of Ba and Kr are higher, hence products are more stable [B1]
This implies that the total binding energy of U is lower than the total binding energies of Ba and Kr. [B1]
Hence energy is released in this fission reaction.

**(iii)**

$$Energy\ released = BE_{products} - BE_{reactants}$$

$$= \left[(141\times 8.3) + (92\times 8.5)\right] - \left[235\times 7.6\right] \quad \text{[M1]}$$

$$= 166.3 = 170 \text{ MeV} \quad \text{[A1]}$$

**7 (a) (i)**

$$\text{vertical velocity} = \frac{\text{change in height}}{\text{duration}}$$

$$= \frac{(7.2-5.5)\times 10^3}{400} \qquad \text{[B1]}$$

$$= 4.25 \text{ m s}^{-1}$$

**(ii)** Since speed is the magnitude of the velocity,

$$v = v_x{}^2 + v_y{}^2$$

$$v_x = \sqrt{v^2 - v_y{}^2}$$

$$= \sqrt{177^2 - 4.25^2} \quad \text{[M1]}$$

$$= 176.95 \quad \text{[A1]}$$

**(b) (i)**

$$\Delta K = \frac{1}{2} m\left(v_f{}^2 - v_i{}^2\right)$$

$$= \frac{1}{2}\times 34000\times(177^2 - 160^2) \quad \text{[M1]}$$

$$= 97393000$$

$$= 9.74\times 10^7 \text{ J} \qquad \text{[A1]}$$

**(ii)** highest power occurs when the aircraft is climbing and speeding up

$$P_{\text{climb}} = \frac{5.67\times 10^8}{400} = 1417545$$

$$P \quad = \frac{9.74\times 10^7}{\ } = 973930 \qquad \text{[M1:Calculating each power]}$$